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DART: De-Anonymization of personal gazetteers through social trajectories

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ABSTRACT

The interest in trajectory data has sensibly increased since the widespread of mobile devices. Simple clustering techniques allow the recognition of personal gazetteers, i.e., the set of main points of interest (also called stay points) of each user, together with the list of time instants of each visit. Due to their sensitiveness, personal gazetteers are usually anonymized, but their inherent unique patterns expose them to the risk of being de-anonymized. In particular, social trajectories (i.e., those obtained from social networks, which associate statuses and check-ins to spatial and temporal locations) can be leveraged by an adversary to de-anonymize personal gazetteers. In this paper, we propose DART as an innovative approach to effectively de-anonymize personal gazetteers through social trajectories, even in the absence of a temporal alignment between the two sources (i.e., they have been collected over different periods). DART relies on a big data implementation, guaranteeing the scalability to large volumes of data. We evaluate our approach on two real-world datasets and we compare it with recent state-of-the-art algorithms to verify its effectiveness.

1. Introduction

Trajectory data consist of ordered sets of spatio-temporal points that mark the coordinates (in terms of latitude and longitude) of moving objects at specific moments in time. The interest towards this kind of data has sensibly increased since the dissemination of mobile devices; indeed, the latter allows to constantly trace the location of the people by registering their connections to cell towers, GPS satellites, and WLAN devices (these are also called *mobility trajectories*). There is enormous value behind trajectory data, as they enable a wide range of *location based services* (LBS), ranging from navigation applications to crowd-sensing applications and target marketing [1]. Their collection and ownership are not limited to infrastructure service providers, but it extends to any application which has been granted access to the geo-localization services of a device.

When trajectory data is ascribed to people, *personal gazetteers* [2] are among the most interesting pieces of information that can be extracted. The personal gazetteer of a user is intended as the set of places mostly visited by such user (e.g., the home and work places, restaurants, shops), possibly enriched with the set of time instants of each visit; we refer to each visited place as a *stay point*. Several research efforts have proven that the discovery of users' personal gazetteers is possible by applying a clustering algorithm on users' trajectories [2]. Due to their sensitiveness, personal gazetteers are usually anonymized; nonetheless, researchers have shown that they present unique patterns (much like fingerprints [3]) that expose them to the risk of being de-anonymized, i.e., of being associated with the individual's true identity. The literature shows that the knowledge of few spatio-temporal points of a user are enough to uniquely identify his/her anonymous trajectory [4,5].

A de-anonymization attack is carried out by an *adversary* to uncover the true identities in an anonymous dataset by exploiting some *external knowledge* where the users' identities are known. To this end, social data is often used as external knowledge due to its relatively easy accessibility [6–8]. Users of social networks often overlook their privacy settings in the platform, leaving their identities and their posts (including the time of posting and the possibly associated georeferenced information, e.g., the location a tweet was posted from) accessible by the general public. A sequence of georeferenced posts on social media still represents trajectory data, namely a *social trajectory*. Although the frequency of points in social trajectories is not very high (and depends on the users' level of interaction with the social network), they can be leveraged by the adversary for the de-anonymization task by measuring its overlapping with the mobility ones.

In this paper, we present DART (*De-Anonymization of peRsonal gazeTteers*), an innovative approach for the de-anonymization of personal gazetteers through social trajectories. The goal is to exploit the

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uniqueness properties of personal gazetteers to measure their similarities with social trajectories. State-of-the-art proposals for de-anonymization of trajectory data follow two main philosophies. The first one is to extract a more-or-less complex behavioral model from both sets of trajectories and to compare them. This approach is hardly applicable to social trajectories, as the sporadic nature of the contained points hinders the extraction of a behavioral model (i.e., social interactions are far less than the number of points available in mobility trajectories). The second approach is to carry out a point-wise comparison, i.e., to calculate a *matching score* between the single points of two given trajectories and to measure their similarity by aggregating these scores. The issue with this approach is that it fails to capture the unique patterns of trajectories, risking random/irrelevant matches to impact significantly on the similarity score. In this context, the contribution of this paper can be summarized as follows.

- We propose DART as an innovative approach that combines the two main philosophies proposed in existing works to properly address the scenario in which social trajectories are used as the external source to de-anonymize personal gazetteers. In particular, it carries out a point-wise comparison to identify the portion of a behavioral model (i.e., the personal gazetteer itself) that is matched by the social trajectory.
- The point-wise matching strategy relies on the computations of a spatio-temporal model from the personal gazetteer, allowing to estimate the location of the user at a certain moment in time. This makes the approach robust to (i) potentially missing data in the mobility trajectories, and (ii) potentially misaligned datasets, i.e., when the mobility and social trajectories are collected over different periods.
- We rely on a big data implementation that guarantees the scalability of the approach to large volumes of data. Indeed, the literature on big data [9,10] often emphasizes the importance of designing scalable and efficient algorithms to enable their applicability on large-scale datasets; while this has also been recognized in the field trajectory anonymization [11,12], it is yet to be addressed by de-anonymization approaches.
- We measure the effectiveness of the approach on two real-world datasets (a proprietary one and the Mobile Data Challenge dataset [13]) and we compare with state-of-the-art algorithms [3,6,8].

The paper is structured as follows: we first discuss the related literature in Section 2; Section 3 introduces personal gazetteers and explains how they are computed, their uniqueness properties and the spatio-temporal model that can be derived; Section 4 presents the details of the de-anonymization approach; the experimental evaluation is discussed in Section 5; finally, the conclusions are drawn in Section 6.

2. Related work

The research community has been paying much attention to the issue of privacy. While some researchers continuously invent new algorithms to effectively anonymize sensitive data without compromising its analytical power [14], others keep discovering ways to break such anonymity [1]. Among the most famous *de-anonymization* attacks is the one by Narayanan et al. [15], where more than 80% of the anonymous users published by Netflix for a competition had been re-identified by matching them with those on the publicly available Internet Movie Database (IMDB). De-anonymization techniques can be applied to very different kinds of data. For instance, some works study social network relationships to infer users' identities [16,17], while others rely on profile data (e.g., username, gender, age) [18,19]. In this section, we focus on approaches that apply to trajectory data.

Many works have underlined how trajectories usually present very unique patterns, such that very few matches are necessary for them to be correctly matched. The work in [20] relies on US census data to demonstrate that home and work locations at the granularity level of blocks is enough to identify almost every American working individual. This work is extended in [21] to generically consider "top N" locations; in this case, the source data consists of Call Details Records (CDRs), i.e., data collected by a cellular operator logging connections of the users to cellular towers. CDRs are considered also in [4], where it is demonstrated that 4 random points in a trajectory are enough to recognize them, i.e., a characteristic resembling fingerprints. When it comes to GPS data (which carry higher accuracy than CDRs) the characterization of trajectories is even sharper, as it is recognized that as little as two points are sufficient to uniquely identify nearly all the users in some datasets [5]. Similar studies have been carried out on social trajectories as well. For instance, [22] evaluates the impact on privacy preservation resulting from the categorization of the locations associated with social media posts. Indeed, the study reveals that the locations carrying the highest discriminating power are the unpopular ones and those that are attached to categories with specific semantics (e.g., "residence").

The research efforts discussed so far mainly warn about the risk of anonymous trajectories being de-anonymized, but do not carry out real de-anonymization attacks. A recent survey [1] has described different approaches that have been proposed to carry-out a de-anonymization attack on trajectory data. An important distinction between them lies in the proposed methodology, which we summarize as *behavioral similarity* and *point-wise matching*.

Behavioral similarity approaches measure the similarity between two trajectories by extracting a behavioral model from both of them and by computing the similarity between the models. One of the most common models is based on the extraction of stay points [23–27]. Freudiger et al. [27] build on [20] to find the matching couples of home and working locations, extracted by applying *k*-means clustering; Manousakas et al. [25] and De Mulder et al. [23] respectively compare the sequence and the graph of the stay points; Naini et al. [24] and Lin et al. [26] compute and compare the histograms of each user's presence in stay points. Another popular model is the Markov chain. Both De Mulder et al. [23] and Gambs et al. [28] propose a method to compare Markov models derived from users' trajectories, while Shokri et al. [29] compute a matching probability between a trajectory and the derived Markov models. The most recent work in this direction is the one by Jin et al. [3], where several *Term Frequency Inverse Document Frequency* (TF-IDF) weighting strategies are evaluated to extract the most relevant points of interest; although the authors focus on the problem of trajectory linking (i.e., given two datasets of trajectories, find the corresponding ones) rather than on de-anonymization, the two problems are equivalent[1]. The issue with behavioral similarity approaches is that they require the adversary's knowledge to be sufficiently rich to build a model that can be compared with the one built from the anonymous source — which is not the typical case. For instance, in the case of social networks, only a few points per user are known to the adversary with respect to the length of the anonymous trajectories. Indeed, these works are all verified by relying on the same dataset to extract both training and test sets of the approach.

Point-wise matching approaches measure the similarity between two trajectories by evaluating the match of the singles points of one trajectory against the other; each match is assigned a *score* or a *probability*, which is then used to compute the final similarity. Cecaj et al. [6] simply count the number of matches between the two trajectories within certain spatial and temporal bounds, while Ma et al. [30] sum the squared error of each match. Other works adopt some statistical information about one of the trajectories to further evaluate each match. For instance, Reiderer et al. [7] extend [30] by incorporating a measure of the importance of encountering a certain user in a certain location and point in time.

[1] De-anonymization can be formulated in terms of entity linking, where the adversary associates anonymous trajectory from a private dataset to its corresponding known trajectory from a known external dataset.

Similarly, Rossi et al. [31] use the frequency of user login in different locations to approximate the probability of visiting these locations. Wang et al. [8] use a Hidden Markov Model to estimate the user's location when the original trajectory is not complete. Differently from behavioral similarity approaches, point-wise matching ones are usually applied to data coming from different sources (e.g., anonymous GPS trajectories, and public social network data). However, the issue with point-wise matching approaches is that they hardly capture the distinguishing features of the trajectories; by aggregating the single matching scores they risk to overvalue matches that occur on less relevant/characterizing points. For instance, a comparison between two trajectories that match several times on meaningless locations (e.g., a stadium during a concert) may return a higher score than the comparison between two trajectories that match few times on significant locations (e.g., home at night).

3. Stay points as personal signatures

A *trajectory* T is a sequence of spatio-temporal points $(t_1, \cdots, t_n)$generated by a moving object (i.e., users in our case), where each point t_i is a triple (*latitude, longitude, timestamp*). In a given source of trajectory data, there exists only one trajectory per user (i.e., a user cannot have more than one trajectory)[2].

The *personal gazetteer* of a user is the set of places that are meaningful to her (e.g., the home, the workplace, a shop frequently visited). These places are also commonly referred to as *stay points* and are typically retrieved by applying some clustering functions on the user's trajectory.

Definition 1. (*Personal gazetteer and stay point*) Given a trajectory T, let $S = \{s_1, \ldots, s_m\}$be the personal gazetteer obtained from T, where s_i is a stay point obtained from T. Each stay point s_i is a triple (*latitude, longitude, Timestamps*) where *latitude* and *longitude* identify the spatial location of the stay point, while *Timestamps* is the set of timestamps for which there exists a point in T that occurs at s_i.

In this section:

- we discuss how stay points are retrieved (Section 3.1);
- we introduce a measure to evaluate the uniqueness of a set of stay points (Section 3.2), which will be used by DART to determine the effectiveness of a match between a social trajectory and a personal gazetteer;
- we present our spatio-temporal model based on the user's personal gazetteer (Section 3.3), which will be used by DART to estimate the user's location in the absence of punctual data.

3.1. Computing stay points

There exist several methods for computing stay points out of trajectory data. Except for some attempts based on exploratory approaches [32] (where the loss of the GPS signal is interpreted as a clue of the user's stopping in a stay point) or k-means clustering [33], most proposals consist of variations of density-based clustering algorithms like DBSCAN [34]. Among the most notable ones, we mention DJ-Cluster [2], a density-joinable clustering technique that improves DBSCAN by adopting optimizations specific to the goal of stay point computation. In particular, DJ-Cluster applies pre-processing rules to the available data to avoid the high memory consumption and the performance decline that would be experienced with DBSCAN [2]. For instance, it focuses only on the points where the user is known to not be moving by discarding every point associated with a speed higher than a certain threshold. Otherwise, the risk would be to include in the identified clusters her most common walking/driving paths.

In this paper, we rely on Patchwork [35], i.e., a density-based algorithm that improves DBSCAN by providing a parallel big data implementation that guarantees linear scalability. Patchwork discretizes the spatial domain into a uniform grid made of 50x50 m cells; the cells with less than 20 points are discarded, the others are aggregated into clusters. Fig. 1 provides the intuition of the algorithm. Noticeably, given a cluster of points, the stay point is chosen as the centroid of the cluster. As the original algorithm is meant for a single trajectory, our implementation is further parallelized to analyze every trajectory in the dataset as a *bag-of-task*. Additionally, we extend Patchwork to include the pre-processing rules introduced by DJ-Cluster. More specifically:

- trajectories with less than 100 points are discarded;
- points with accuracy greater than 30 m are discarded;
- points with an associated speed higher than 2 km/h are discarded;
- each trajectory is sampled down to 1 point per minute to avoid the generation of clusters on random stopping points (e.g., due to traffic lights).

3.2. Uniqueness of stay points

Stay points alone are often enough to uniquely identify a user among the others [4,20,21]. Consider $\mathcal{S} = \{S_1, \cdots, S_n\}$the set of available personal gazetteers and $W = \{s \mid s \in S, S \in \mathcal{S}\}$the universe of all stay points. Then, let us define the concept of adjacency between stay points as follows.

Definition 2. (*Stay points adjacency*) A stay point $s_1 \in W$ is *adjacent* to a stay point $s_2 \in W$ (i.e., $s_1 \cong_{adj} s_2$) if the spatial distance between the two stay points is lower than a spatial threshold ω, i.e., $dist_{space}(s_1, s_2) \leq \omega$.

Clearly, we are interested in evaluating the adjacency between stay points belonging to different personal gazetteers. In particular, the adjacency definition is necessary to introduce a concept of uniqueness: we say that $s_1 \in W$ is *unique* if there is no adjacent stay point $s_2 \in W$ that belongs to a different personal gazetteer.

Definition 3. (*Stay point uniqueness*) A stay point $s_1 \in W$ is *unique* if $\nexists\, s_2 \in W$ such that $s_1 \cong_{adj} s_2$, $s_1 \in S'$, $s_2 \in S''$, $S' \neq S''$.

Consider the example in Fig. 2, where the stay points of three personal gazetteers are plotted; it is $S' = \{s_1, s_2, s_3\}, S'' = \{s_4, s_5\}, S''' = \{s_6, s_7\}$. The stay points of S' are not individually unique, because each of them is adjacent to stay points of other personal gazetteers. However, their combination represents a unique signature. More specifically, the subsets $\{s_1, s_3\}$ and $\{s_2, s_3\}$ are also unique, because S' is the only personal gazetteer that can be traced back to both places in both cases.

Definition 4. (*Personal gazetteer subset uniqueness*) Let S^* be a subset of a personal gazetteer $S' \in \mathcal{S}$; we define the *uniqueness*of S^* in $\mathcal{S}$as an inverse function of the number of personal gazetteers $S \in \mathcal{S}$such that every stay point in S^* is adjacent to some stay points in S.

$$unq(S^*) = \frac{1}{\left|\left\{S \in \mathcal{S} \mid \forall s_1 \in S^* \ \exists s_2 \in S, s_1 \cong_{adj} s_2\right\}\right|}$$

Notice that $\mathcal{S} \supseteq S'$, thus the denominator is always greater than 1. This concept of uniqueness is used in DART to effectively verify the de-anonymization of a user: the best-case scenario is when a social trajectory matches a unique subset of a personal gazetteer. To this end, the adjacency threshold ω plays an important role: on the one side, the higher the value, the higher is the importance given to stay points in remote locations; on the other side, a low-value risks to consider all stay points as unique. In any case, the adjacency threshold should be based on an empirical evaluation of the datasets and consider the accuracy of stay points' coordinates. In our datasets, the coordinates of stay points

[2] In other works, trajectories are bounded by a proximity constraint, i.e., the time difference between a given t_i and $t_{i+1}$is always lower than some threshold; this constraint is not considered in this work.

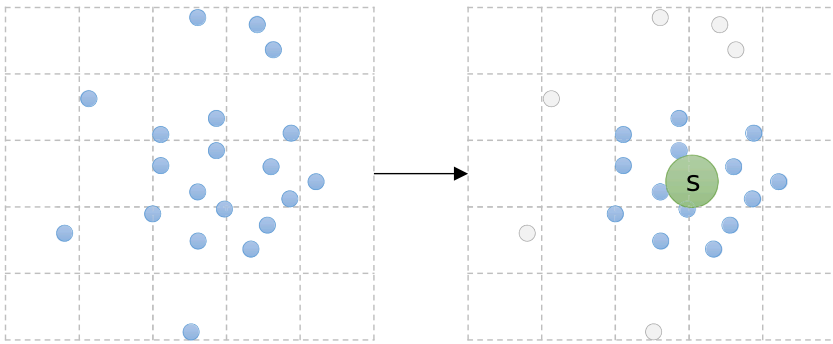


Fig. 1. Intuition of the algorithm for stay point extraction.

Fig. 2. Example of stay points of three personal gazetteers.

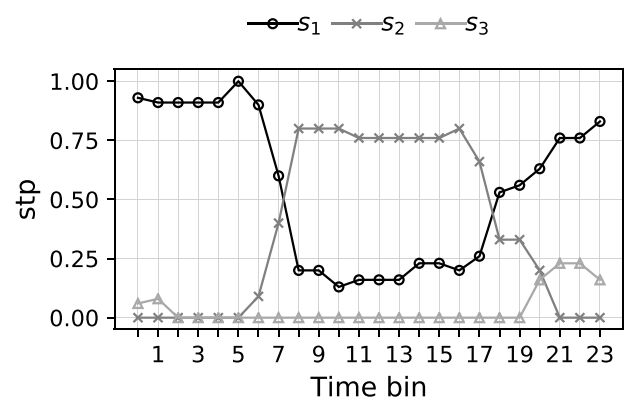


Fig. 3. Example values of *stp* on three stay points s_1 (home), s_2 (work), and s_3 (pub/restaurant) in the different time bins.

are obtained by locating the centroids of clusters of GPS points (see Section 3.1); thus, such coordinates are not exact and entail a certain degree of approximation, which depends on how the GPS points are distributed within the cluster. A measure of this approximation is given by the average diameter of the aforementioned clusters, which we measure as 200 m in our datasets. Based on this evaluation, we set ω at 200 m.

3.3. Stay point spatio-temporal model

In this work, we exploit stay points to define a spatio-temporal model that enables the estimation of a user's location in a certain moment. This is necessary to "fill-in the blanks", as it is not uncommon for anonymous trajectories to be incomplete (e.g., the GPS trace may be recorded only under certain conditions) or for social trajectories to be misaligned with anonymous ones (e.g., because the recording period is different). The granularity we consider is the one of the *hour of the day*, i.e., we split time into 24 time bins[3]. Let S be the personal gazetteer of a certain user, $s \in S$ a stay point of the personal gazetteer, and $b \in [0, 23]$ a time bin. Then, we define $stp(s, b)$ as the empirical probability of the user being in stay point s in time bin b.

Let $timeBin : \mathbb{N} \rightarrow [0, 23]$ be the function that returns the time bin corresponding to a certain timestamp. Also, let $days(s, b) \in [0, \mathbb{N}]$ be number of distinct days along the monitoring period for which there exists a timestamp ts in s such that $timeBin(ts) = b$. Then, the empirical *spatio-temporal probability* of the anonymous user being in s during time bin b is:

$$stp(s, b) = \frac{days(s, b)}{\sum_{s_n \in S} days(s_n, b)}$$

Intuitively, $stp(s, b) \in [0, 1]$ indicates the percentage of days for which the user has been recorded being in s and not in other stay points. For instance, in the typical case, $stp(s, b) \rightarrow 1$ for the stay point representing the home place during night hours (i.e., $b \in [1, 5]$). Fig. 3 shows some examples for the values of *stp* on three stay points s_1 (resembling a home place), s_2 (resembling a work location), and s_3 (resembling a pub or restaurant) in the different time bins.

4. The de-anonymization approach

The de-anonymization approach relies on anonymous personal gazetteers, each of which represents a *signature* of a user's most frequented places. Indeed, a personal gazetteer (or even a subset of it) is often unique within a population of users [20,21,27]. Let us define with $\mathcal{S}$ the set of anonymous personal gazetteers and with $\mathcal{K}$ the set of social trajectories (e.g., geo-localizated tweet histories)[4]; the goal of the approach is to associate each social trajectory $K \in \mathcal{K}$ to the most probable personal gazetteer $S \in \mathcal{S}$. To this end, we proceed in two steps.

1. The first step consists in calculating a *global score* between each social trajectory K and personal gazetteer S. The global scoring function relies on *local scores* that are calculated by measuring the spatio-temporal distance between the single points $k \in K$ and the single stay points $s \in S$. The details of this step are discussed in Section 4.1 and Section 4.2.

[3] Time bin 0 refers to the time slot between 00:00 and 00:59; time bin 23 refers to the time slot between 23:00 and 23:59.

[4] The letter "K" refers to the social trajectories being *known* instead of anonymous.

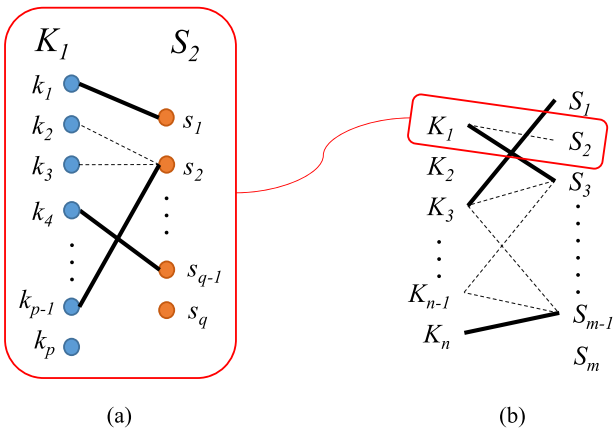


Fig. 4. On the left (a) the scoring step between a social trajectory K_1 and a personal gazetteer S_2; edges represent local scores, and the bold ones are those used to compute the global score. On the right (b) the de-anonymization step to assign to each social trajectory $K \in \mathcal{K}$its most probable personal gazetteer $S \in \mathcal{S}$; here, edges represent global scores, and the bold ones represent the chosen assignments.

2. The second step consists in assigning each K to the most probable S based on the computed global scores. Its details are discussed in Section 4.3.

Fig. 4 shows an intuition of the two steps. In the scoring step (i.e., the left figure), each point k is associated with the spatio-temporally closest stay point s; the edges represent these associations. Then, the global score is based on the maximum scores for each stay point (i.e., the bold edges); indeed, the more stay points in S are matched by the points in K, the more significant is the correspondence between K and S. In the de-anonymization step (i.e., the right figure), the rationale is similar: among all the possible matches between personal gazetteers and social trajectories (i.e., the edges), we find the best matches (i.e., the bold edges) such that (a) each K is matched to up to one S, and (b) each S is matched to up to one K. This is necessary to avoid having the same social trajectory being assigned to multiple personal gazetteers, and vice versa.

4.1. The local scoring function

In order to evaluate the match between a social trajectory and a personal gazetteer, we must first define a local scoring function to evaluate the single matches between the points $k \in K$ and the stay points $s \in S$. Given a couple (k, s), the local score builds on two principles: (1) the *closeness* between k and s from both spatial and temporal perspective; (2) the *uniqueness* of k with respect to S, which depends on whether k is close to other stay points belonging to different personal gazetteers (i.e., other than S).

Let us start from the spatio-temporal closeness between a point k and a stay point s. Measuring the spatial correspondence is trivial, as both are associated with a spatial location in terms of latitude and longitude. Conversely, to measure the temporal correspondence we rely on the set of known timestamps for s, which indicate the moments in time when the anonymous user has been recorded in s. However, due to the potential incompleteness of trajectory data, it may be hard to find a punctual correspondence. In the absence of the latter, we rely on the spatio-temporal model of the personal gazetteer (Section 3.3) to measure the probability of the anonymous user being in s around the time of k. Thus, we define the closeness measure as a function of three components, namely Δ_{space}, Δ_{time}, and Δ_{habit}. The first two components rely on folded Gaussian models, which are often used to represent probability density functions[5]; thus, they allow measuring the matching probability between two spatio-temporal points. Consider $I = \{k \mid k \in K, K \in \mathcal{K}\}$the universe of all known points and $W = \{s \mid s \in S, S \in \mathcal{S}\}$the universe of all stay points.

- Let Δ_{space}: $(I, W) \rightarrow [0, 1]$ be the function defined by

  $$\Delta_{space}(k, s) = g\left(dist_{space}(k, s) \,/\, 200\right)$$

  to calculate the *spatial closeness* between k and s. Here, $dist_{space}(k, s)$ is the spatial distance in meters between k and s, and g is a folded Gaussian distribution with mean $\mu = 0$and standard deviation $\delta = 0.4$. This ensures that $\Delta_{space}(k,s) = 1$when $dist_{space}(k,s) = 0$and that $\Delta_{space}(k, s) \approx 0.5$ when $dist_{space}(k, s) = 100$meters. Note that the normalizing factor of 200 meters is consistent with the threshold for stay point adjacency ω set in Section 3.2.
- Let Δ_{time}: $(I, W) \rightarrow [0, 1]$ be the function defined by

  $$\Delta_{time}(k, s) = g(dist_{time}(k, ts) \,/\, 3600)$$

  to calculate the *temporal punctual closeness* between k and s. Here, ts is the timestamp in s that is closest to the timestamp in k, $dist_{time}(k, ts)$ is the temporal distance in seconds between k and ts, and g is defined as above. This ensures that $\Delta_{time}(k,s) = 1$when $dist_{time}(k,ts) = 0$and that $\Delta_{time}(k, s) \approx 0.5$ when $dist_{time}(k, ts) = 30$minutes. The normalizing factor of 1 h (i.e., 3600 seconds) is consistent with the granularity of time bins in our spatio-temporal model (Section 3.3) and with the granularity considered by several related approaches (e.g., [3,8]).
- Let Δ_{habit}: $(I, W) \rightarrow [0, 1]$ be the function defined by

  $$\Delta_{habit}(k, s) = \sum_{(x,y)\in\gamma} stp(s, b + x) * y$$

  to calculate the *temporal behavioral closeness* between k and s. In practice, Δ_{habit} is the weighted sum of the empirical probabilities of the user being in s at different time bins: b is the time bin of k, and $\gamma = \{(-2, 0.1), (-1, 0.2), (0, 0.4), (+1, 0.2), (+2, 0.1)\}$is a set of (x, y) couples where x is a time bin index relative to b, and y is a weight to be applied to the stp function[6]. This means that $stp(s, b)$ is weighted 0.4, $stp(s, b+1)$is weighted 0.2, and so on. The values of γ have been obtained from empirical evaluations, but they can be tuned as long as $\sum_{(x,y)\in\gamma} y = 1$.

  The intuition is to accommodate behavioral evolution by considering not only the empirical probability at the exact time bin b(i.e., $stp(s, b)$) but also the probabilities in the closest time bins. For instance, with reference to Fig. 3, consider a point k made in time bin $b = 5$; here, $\Delta_{habit}(k, sp_2) = stp(sp_2, 3) * 0.1 + stp(sp_2, 4) * 0.2 + stp(sp_2, 5) * 0.4 + stp(sp_2, 6) * 0.2 + stp(sp_2, 7) * 0.1 = 0 + 0 + 0 + 0.02 + 0.04 = 0.06$. Thus, instead of completely refusing the possibility of the user being in sp_2at the time bin of k (as $stp(sp_2, 5) = 0$), we admit a small probability (as the empirical evidence shows the user being in sp_2in the immediately following time bins).

Given the aforementioned components, we define the spatio-temporal closeness as follows.

Definition 5. (*Closeness*) The closeness between a point k and a stay point s is:

[5] Gaussian models have been used in the literature to measure closeness between points, not only in de-anonymization approaches (e.g., [8]) but also in other contexts (e.g., [36]). In particular, we rely on a folded Gaussian model as our input values are always non-negative.

[6] We assume circularity in the time bins: e.g., $23 + 2 = 1$, and $0 - 2 = 22$.

$$closeness(k,s) = \Delta_{space}(k,s)\cdot(\Delta_{time}(k,s) + (1-\Delta_{time}(k,s))\cdot\Delta_{habit}(k,s))$$

In practice, Definition 5 measures closeness as the product of spatial closeness and temporal closeness[7]. Spatial closeness is simply determined by Δ_{space}, while temporal closeness is composed by the punctual and the habit components. The priority is given to the punctual component (i.e., Δ_{time}), while the weight of the habit component (i.e., Δ_{habit}) is the inverse of the punctual one. Indeed, timely correspondences convey more precise information than those based on the users' habits. Example 1 further explains how temporal closeness is obtained.

Example 1. The mechanism of the temporal component is further explained by the following scenarios.

- $\Delta_{time} = 1$: there is a punctual correspondence of the anonymous user being in $s$at the time of k; thus, independently of the user's habits Δ_{habit}, the temporal component measures 1.
- $\Delta_{time} = 0$: there is no punctual correspondence of the anonymous user being in $s$at the time of k; thus, the temporal component depends on whether there is any probability of the user's being in s during k's time bin based on its habits Δ_{habit}.

The closeness measure defines how well a point k matches a stay point s, but it does not capture the potential crowding that is contextual to this match. Indeed, if different anonymous users are spatio-temporally close to k, the latter will obtain a high closeness to the stay points of the former's personal gazetteers. To this end, we introduce the point uniqueness[8] as a measure of the actual discriminating power of a point k with respect to S.

Definition 6. (*Point uniqueness*) The uniqueness of a point k with respect to S is:

$$unq(k,S) = \frac{\sum_{s\in S} closeness(k,s)}{\sum_{S'\in\mathcal{S}}\sum_{s'\in S'} closeness(k,s')}$$

The intuition is that the closer k is to stay points of different personal gazetteers, the lower will be its discriminating power with respect to S. Notice that k may be close to more than one stay point belonging to the same personal gazetteer; this is why the point uniqueness with respect to S positively considers the closeness with all the stay points in S.

Ultimately, we define the local score of a couple (k, s) by combining the measures of closeness and point uniqueness.

Definition 7. (*Local score*) The local score of a couple (k, s) is:

$$localScore(k,s) = closeness(k,s)\cdot unq(k,S)$$

4.2. The global scoring function

The global score quantifies how well a social trajectory K matches a personal gazetteer S. Let us start by defining the mapping of K into S by means of a *mapping signature*, which associates each point $k \in K$ to the spatio-temporally closest stay point $s \in S$.

Definition 8. (*Mapping signature*) Given a social trajectory K and a personal gazetteer S, the mapping signature represents the mapping of K into S as set of couples $M = \{(k_1,s_1),\ldots,(k_n,s_n)\}$. It is obtained through a partial function φ: $K \rightarrow M$that maps each point $k \in K$ to the stay point $s \in S$ that maximizes the local scoring function *localScore*(k, s) (Section 4.1). Formally, $\varphi(k) = \text{argmax}_{s\in S} localScore(k,s) \mid localScore(k,s) > 0$.

The mapping function φ is partial as a point $k \in K$ may be too spatio-temporally far from any of the stay points in S — and the same holds the other way around. The global score between K and S is calculated on the obtained mapping signature M.

Definition 9. (*Global score*) Let us refer to K_M and S_M as the set of points and of stay points that belong to the mapping signature M, respectively. Also, for a given $s \in S_M$, consider $k^* \in K_M$ the point mapped to s with the highest local score, i.e., $k^* = \text{argmax}_{k\in K_M} localScore(k,s)$. Then, the *global score*of the mapping signature M is:

$$globalScore(M) = unq(S_M)\cdot\sum_{s\in S_M} localScore(k^*,s)$$

Whereas other works (e.g., [6]) simply aggregate the local scores, our global scoring function is based on a qualitative and a quantitative principle. The first part of the equation (i.e., the multiplication by the uniqueness of the matched stay points, which is defined in Section 3.2) evaluates the quality of the mapping in terms of the discriminating power of the matched stay points: the higher their uniqueness, the higher is the final score. The second part of the equation (i.e., the sum of closeness values) evaluates the quantitative aspect of the mapping by summing the local scores. More specifically, we consider only the maximum local score for each matched stay point in order to favor the mappings that cover a higher number of stay points and, therefore, are more representative of the personal gazetteer.

4.3. The assignment step

Following the calculation of the global score between each couple in ($\mathcal{K}$, $\mathcal{S}$), we are left with the problem of assigning to each Kits most probable match S. As anticipated, we want to avoid the assignment of the same K to multiple personal gazetteers (and vice versa) and to maximize the chosen global scores. This problem is well known in the literature as the assignment problem [37]. Let $G = (\mathcal{K}, \mathcal{S}, E)$be a weighted bipartite graph where the edges E correspond to the global scores between the social trajectories and the personal gazetteers. Then, the assignment simply consists in finding the maximum-weight matching.

Finally, DART's approach is summarized in Algorithm 1. Notice that the computation of the global score between a known trajectory K and a personal gazetteer S cannot be modeled as a simple function of K and S; this is because the computation of point uniqueness (which is a component of the local score) requires a global knowledge of the closeness values between every point k and every stay point s. Thus, we start by computing closeness values between every point k and every stay point s (function *computeCloseness*, Line Algorithm 1), then we

Input $\mathcal{S}$: personal gazetteers; $\mathcal{K}$: social trajectories; φ: mapping function.
Output matches between personal gazetteers and social trajectories.
1: $C \leftarrow computeCloseness(k,s)\forall(k,s) \in (\mathcal{K},\mathcal{S})$
2: $U \leftarrow computePointUnq(k,S,C)\forall(k,S) \in (\mathcal{K},\mathcal{S})$
3: $E \leftarrow \varnothing$
4: **for all** $(K,S) \in (\mathcal{K},\mathcal{S})$ **do**
5: $\quad L \leftarrow computeLocalScore(k,s,C,U)\forall(k,s) \in (K,S)$
6: $\quad M \leftarrow computeMapping(K,S,L,\varphi)$
7: $\quad E \leftarrow E \cup \{K,S,computeGlobalScore(M,L)\}$
8: **return** $computeAssignment(\mathcal{K},\mathcal{S},E)$

Algorithm 1. DART algorithm.

compute point uniqueness values between every point k and every personal gazetteer S (function *computePointUnq*, Line 2). At this point, we can iterate through the (K, S) couples to determine local scores, mappings and global scores (Lines 4 to 7). The *computeAssignment*

[7] Notice that many de-anonymization approaches evaluate the closeness between two points in terms of both spatial and temporal components (e.g., [6,8,30,31]).

[8] We refer to this measure as *point uniqueness* in order to differentiate it from the *uniqueness* introduced in Section 3.2, which applies the same principle to sets of stay points.

Table 1
Statistics about the MIL and MDC datasets.

Parameter	MIL	MDC
Number of users	100	190
Avg number of pings per user (anonymous)	2040	13,170
Avg number of pings per user (social)	22	9123
Avg number of stay points per user	5	21
Avg number of days per user	28	185

function (Line 8) finally computes the maximum-weight matching on the weighted bipartite graph.

5. Evaluation

Two anonymous datasets are used in this work, whose statistics are reported in Table 1. The first one is a proprietary, real-world dataset based on a case study scenario centered on (but not limited to) the Italian city of Milan; for brevity, we will refer to this dataset as MIL. It spans from September 2017 to December 2017 and includes nearly 500,000 anonymous GPS trajectories. For the sake of simplicity, we extract from this dataset a sample of 100 trajectories of users whose personal gazetteers indicate that they live and work in Milan. As we do not possess ground-truth data of social trajectories, we simulate the latter by generating them from the anonymous ones. Each generated trajectory contains a random 1% sample of the original one[9], where each point is injected with a spatial and a temporal jitter, misplacing it of up to 100 m and up to 30 minutes. We remark that (i) the spatial jitter is injected to the original GPS points (not to the calculated stay points), meaning the generated social points can easily fall quite distant from stay points; (ii) as time bins are constant (e.g., 00:00-00:59, 01:00-01:59), a 30 minutes difference is likely to put the social point into a different time bin than the respective GPS point.

Different scenarios are simulated by generating different sets of social trajectories. First, we vary the percentage of points located at a stay point (i.e., whether people interact more on social networks while commuting or while at a stay point); we consider a point p to be located at a stay point s if $dist_{space}(p, s) \leq 100$ meters. Also, we vary the temporal misalignment between the social and the anonymous trajectories. The second dataset is the Mobile Data Challenge (MDC) dataset [13], which provides real-world data of nearly 200 individuals from the Lake Geneva region (Switzerland) in 2010–2011. For each user, it provides a sample of his/her personal gazetteer (which we will use as ground truth to verify the stay points inferred by DART), as well as the trajectories captured by GPS satellites and WLAN devices in a complementary way (i.e., either one of them is active for a single user at the same time, not both). In this case, we use the GPS satellite as the source of anonymous trajectories and the WLAN as the source of the social ones.

MIL is by far the most challenging dataset due to the higher spatio-temporal density of the trajectories. In particular, GPS trajectories in MDC are often disjoint from the temporal perspective. For this reason, our evaluation mostly focuses on the former dataset. Nonetheless, the unusual characteristic of MDC (i.e., the complementarity of GPS and WLAN sources) allows us to further evaluate the robustness of the approach in a different scenario. Additionally, we remark that MDC is a freely-available real-world dataset that provides a solid basis for testing and comparing de-anonymization approaches.

A prototypical implementation of DART has been realized on a big data infrastructure, consisting of 18 Ubuntu machines mounting the Cloudera Distribution for Hadoop (CDH) 6.2.0; the source code is available on GitHub[10]. The datasets are stored as Parquet tables on Apache Hive, while the approach is implemented on Apache Spark.

5.1. Stay points validation

In Section 3 we have briefly discussed the Patchwork algorithm that we adopt to infer stay points from GPS trajectories. In this Section, we aim to verify the correspondence between the stay points that we infer and the ground truth provided by the MDC dataset — even though it is only a sample of the user's personal gazetteers. In MDC, stay points are simply called *places*; some of them have semantic labels attached (e.g., distinguishing home places from working locations and other kinds of stay points), but none of them contains spatial information. However, since the dataset provides the starting and ending timestamps of each visit to these places, we are able to estimate their location by averaging the latitude and longitude recorded for the user between such timestamps.

The MDC sample contains 493 places (i.e., 2.6 for each user on average), whereas our stay point extraction algorithm infers 3714 (i.e., 19.5 for each user on average). We have verified that 84% of MDC places have a stay point in its close vicinity (i.e., no more than 200 m apart). Our implementation of Patchwork fails to recognize the remaining places due to either the low volume or the sparsity of geolocated records, which hinders the recognition of a cluster. Conversely, only 11% of our stay points can be traced back to MDC places, meaning that 89% are new with respect to the data provided by MDC.

5.2. Stay points uniqueness

The fundamental assumptions in DART are that (1) users' social interactions mainly happen on stay points, and (2) personal gazetteers are often unique and enable the de-anonymization of users. Fig. 5 verifies the latter statement in the MIL and MDC datasets: given a certain percentage of stay points from each user's personal gazetteer, it shows the average probability that such subset is unique within the population (Definition 4 in Section 3.2). The Figure not only confirms that the personal gazetteer is indeed a unique signature, but it also indicates that even half of it is enough to identify about 80% of the population in MIL. The easier de-anonymizability of MDC is also confirmed, as only one third of stay points are enough to identify 80% of the population. Ultimately, we conclude that we can effectively rely on stay points to de-anonymize trajectories.

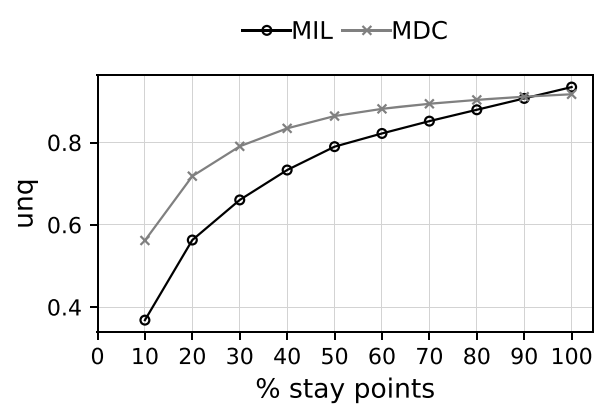


Fig. 5. Uniqueness probability of subsets of stay points from personal gazetteers.

5.3. Efficiency of DART

State-of-the-art de-anonymization approaches usually focus on the effectiveness of the algorithm and overlook the evaluation of its efficiency, even though the importance of designing scalable and efficient algorithms in order to enable their applicability on large-scale datasets is

[9] The number of social interaction is usually quite low when compared to the frequency of GPS signals.
[10] https://github.com/big-unibo/dart

often emphasized in the literature [11,12]. We implemented DART on Apache Spark in order to guarantee the scalability of the approach to large volumes of data. To evaluate DART's efficiency, we measure its execution time on MIL by progressively increasing the number of users considered in the dataset (from 100 to the whole 500,000) and compare it with the sequential implementations (in Python) of Cecaj et al. [6], Jin et al. [3], and Wang et al. [8][11].

The configuration of Spark for DART's execution consists of 10 executors with 3 cores and 10GB of RAM for each executor. For the sake of a fair comparison, we parallelized the algorithms of [6] and [8] as bags-of-tasks and ran 30 concurrent instances that independently worked on 1/30th of the dataset[12]. Conversely, the algorithm of [3] cannot be parallelized (in fact, it requires the whole dataset to run its TF-IDF strategy to compute the signatures of trajectories), thus it has been executed on one of the available machines.

The results are shown in Fig. 6; we remark that both axis are logarithmic and that executions that took more than a day have been halted and are shown as full-sized bars. The parallel nature of DART's implementation entails a slight disadvantage when working on small datasets (due to the overhead for managing distributed computations), but it greatly rewards when the dataset's size becomes significantly large. [8] immediately reaches an exaggerate execution time, even with a medium-sized dataset. This is probably due to its complex algorithm, which relies on a Hidden Markov Model to fill incomplete trajectories by enumerating all possible complete trajectories; although the algorithm allows [8] to achieve fairly good results (as we will show in Section 5.4), it sensibly hinders its application on large-scale datasets. As for [6], it reaches high execution times as well, with a linear increase with respect to the increase of trajectories. In particular, the execution time in [6] is mainly given by the pairwise comparison of the point of every couple of trajectories, whereas we use spatial joins to compute local scores only between points that are close enough. The execution time of [3] follows a pattern similar to [6]. Considering that [3] uses 1/30th of the computation power, it is actually considerably fast; indeed, its computation time is mainly due to the generation of trajectories' signatures, which are easily compared once they have been computed. However, as we will show in Section 5.4, [3] trades a faster computation with a low accuracy in almost every scenario.

5.4. Effectiveness of DART

In this section, we evaluate the effectiveness of the de-anonymization approach on both datasets by means of the *accuracy*, defined as the fraction of correctly guessed assignments over the actual number of users. The results are compared with [3,6,8].

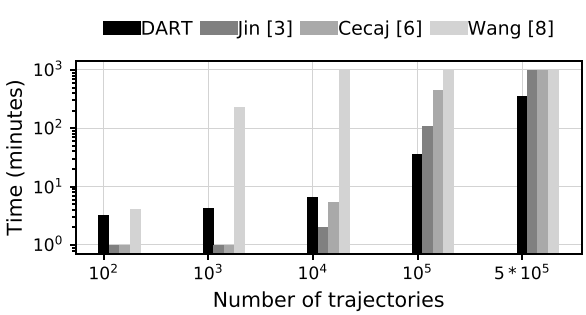


Fig. 6. Efficiency of DART and other de-anonymization approaches on the MIL dataset considering a progressively increasing number of trajectories.

[11] To the best of our knowledge, [3] and [8] are the most recent related approaches in the literature.

[12] Differently from DART, the computation of the score between every (K, S) in [6] and [8] is independent of the others.

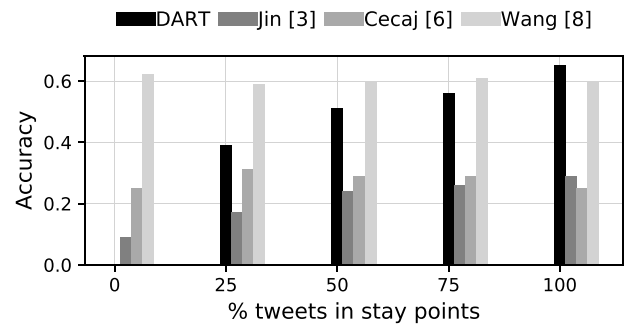
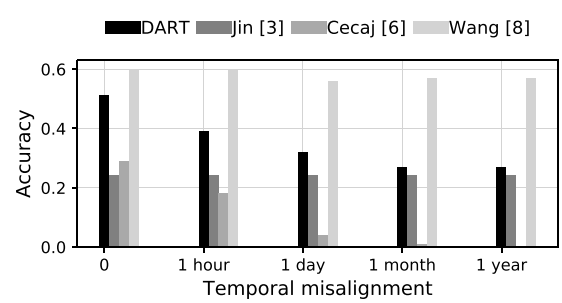
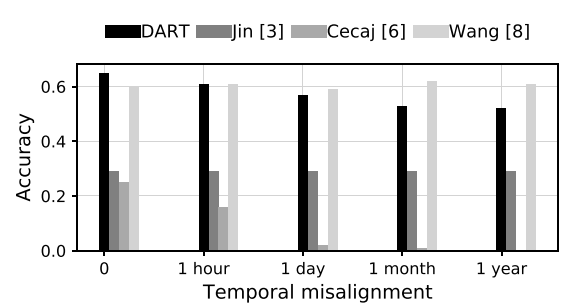


Fig. 7. Accuracy of DART and other de-anonymization approaches on the MIL dataset.

In the first experiment, we have created several sets of social trajectories on MIL by varying the percentage of points located at a stay point. Considering the scenario of social trajectories being Twitter histories, the extremes are that users always tweet while traveling (i.e., 0% of points within a stay point) and that users always tweet from home, work, or other stay points (i.e., 100% of points within a stay point). The results are shown in Fig. 7. Except for the case where points in K always fall outside of stay point (i.e., what we assume to be a very unlikely scenario), DART performs better than [6] and [3]. In the former scenario, the poor performance of DART is physiological, as points outside of stay points are not considered. Indeed, the results also show that DART's accuracy (as well as the accuracy of [3]) increases as the percentage of points in K located at stay points also increases (whereas [6] is unaffected by such variation). Interestingly, the fact that the accuracy of [3] (which uses a TF-IDF strategy to summarize each trajectory) follows the same trend of DART confirms that stay points are signatures traceable to unique users; therefore, we conclude that the more data is located at stay points, the easier it is to de-anonymize it. Ultimately, the comparison with [8] shows that the latter achieves consistently good accuracy, comparable to the one of DART in the latest scenarios. Indeed,



(a) The temporal misalignment starts from the 50% configuration in Figure 7.



(b) The temporal misalignment starts from the 100% configuration in Figure 7.

Fig. 8. Accuracy of DART and other de-anonymization approaches on the MIL dataset considering a progressive temporal misalignment between personal gazetteers and social trajectories.

[8] adopts a complex technique based on a Hidden Markov Model to estimate the user's location when the original trajectory is not complete. However, as shown in Section 5.3, the good accuracy of this approach comes at the expense of computational complexity, which hinders its adoption on large-scale datasets.

A second experiment verifies the robustness of DART to the temporal misalignment between social trajectories and personal gazetteers. Starting from the 50% and 100% configurations of the previous experiment, we simulate the misalignment by adding a certain temporal delta to every point in the social dataset. The results are shown in Fig. 8. As in the previous experiment, [8] is able to maintain high accuracy, even though its execution times remain severely high. DART's accuracy is actually affected by the temporal misalignment, but the behavioral component of DART is able to maintain the former always above 27% and 52% in the respective scenarios. This demonstrates the robustness of DART, as stay points are proven as a solid bedrock for the de-anonymization. The accuracy of [3] is constant because their algorithm does not consider the temporal dimension; although this characteristic makes the approach robust to temporal misalignment as well, it prevents [3] from achieving better accuracy when the temporal dimension is more relevant. Ultimately, the accuracy of [6] progressively decreases down to zero; this is inevitable, as [6] only looks for punctual temporal matches.

Finally, we evaluate the accuracy of DART on MDC. On this dataset we observe a significant difference between the four approaches: as shown in Fig. 9, DART is able to achieve an optimum result (i.e., a 91% accuracy) and performs far better than the other algorithms. The poor performance of [6] is due to its algorithm, which abruptly excludes a match between K and A if there exists a record of A being too far from K (i.e., $dist_{space}(a, k) > 2 \cdot \tau_s$) in a close time windows (i.e., $dist_{time}(a, k) \leq \tau_t$). This is called the *exclusion rule* in [6], and it is based on the principle that a person cannot be in two places at the same time. However, the complementary nature of the original GPS and WLAN trajectories in MDC makes it easier for the exclusion rule to be activated. Conversely, the good performance of DART and [3] (which is much higher than the one on MIL) is due to the trajectories in MDC being less overlapped; in fact, all trajectories in MIL are bounded within the city of Milan over one month, whereas in MDC they are centered on the Geneva lake and can span up to Zurich over one year (i.e., a much larger spatio-temporal area). Interestingly, [8] is unable to achieve a better accuracy than the one reached in MIL. This is probably due to the complementarity of GPS and WLAN sources (mentioned at the beginning of this Section): the behavior of GPS and WLAN trajectories in MDC is quite different, because the former mostly captures movements between stay points, while the latter captures visits to stay point. To this end, MDC is similar to the last scenario in MIL where tweets are mostly made within stay points (Fig. 7). We conclude that [8] achieves a consistent accuracy in different scenarios, but it is unable to exploit the one in which the concentration of social interactions is mostly around stay points (which, as we recall, is one of our main assumptions).

6. Conclusions and future work

In this paper, we proposed an original approach to trajectory de-anonymization that relies on personal gazetteers to model users' behaviors and to effectively capture and recognize each user's spatio-temporal signature. Our experimental evaluation has proven that DART improves recent state-of-the-art algorithms by effectively coupling an efficient implementation with high effectiveness on different datasets and scenarios. Our algorithm is also robustness to temporal misalignment between the anonymous and known trajectories. The only related approach that achieves better results in some scenarios is [8], whose elevated complexity severely hinders its applicability on large-scale datasets. At this stage we are currently working in different directions to further improve the effectiveness of the approach by: (a) enhancing the behavioral model to improve its effectiveness in case of

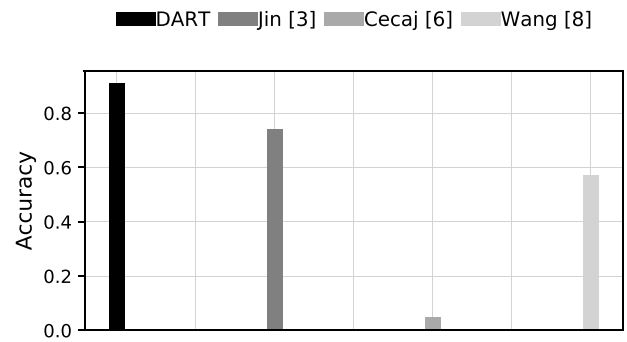


Fig. 9. Accuracy of DART and other de-anonymization approaches on the MDC dataset.

significant temporal misalignment, e.g., by considering not only the hour of the day but also the day of the week; (b) comparing it with other state-of-the-art algorithms, to improve our understanding as to which scenarios are better handled by different algorithms, with particular focus on those scenarios where [8] achieves higher accuracy. To this end, we will investigate whether it is possible to further improve our accuracy without burdening the efficiency of the algorithm; (c) collecting — possibly with the help of volunteers — a large-scale ground-truth dataset to validate the goodness of the approach; (d) measuring the robustness of DART against anonymization algorithms (e.g., k-anonymity [38], differential privacy [39]), which hinder the de-anonymization of trajectory data by either removing sensible spatio-temporal points or by decreasing the precision of the latter.

CRediT authorship contribution statement

Matteo Francia: Conceptualization, Methodology, Validation, Writing - review & editing. **Enrico Gallinucci:** Conceptualization, Methodology, Software, Writing - original draft. **Matteo Golfarelli:** Supervision, Conceptualization, Methodology, Writing - review & editing. **Nicola Santolini:** Investigation, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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